

ÉquiAlgo: fair student financing

Engineering and Computer Science Hackathon 2026. 24-hour challenge. | Participant brief

HxBuddy only displays indicative F1 and accuracy metrics for this challenge. It does not calculate the 35 points in the official rubric below. IVADO evaluates the final deliverables and that rubric.

A Quebec financial institution scores scholarship and student-loan applications with a machine learning model. It is 88% accurate. An internal audit found it grants awards to 48.4% of applicants from Montréal and the Capitale-Nationale, against 27.3% from Bas-Saint-Laurent, Côte-Nord and Gaspésie-Îles-de-la-Madeleine.

Average R score is 27.3 in the remote regions and 28.0 in the centres. That accounts for part of the 21-point gap. The rest is unexplained.

Your task: diagnose the bias, correct it, and propose a monitoring plan for production.

All data is synthetic. The institution is fictional.

Setup

```
git clone <YOUR_REPO_URL>
cd defi-equalgo
python3 -m venv venv
source venv/bin/activate      # Windows: venv\Scripts\activate
pip install -r requirements.txt
jupyter notebook baseline_model.ipynb
```

Python 3.10 or newer. Run the notebook once before changing anything. It trains the production model, measures it, and audits it.

Files

File	Rows	Contents
data/donnees_demandes.csv	10,000	historical applications, with decision_octroi
data/candidats_evaluation.csv	4,000	applications to score, no label
baseline_model.ipynb		production model, metrics, fairness audit

Columns

Column	Meaning
id_candidat	identifier, C000 000 format
cote_r_equivalent	academic performance, R score equivalent, 15 to 40
programme_etudes	program of study, 5 categories
region_administrative	sensitive attribute, Quebec administrative region
code_postal_3	first three characters of the postal code
revenu_familial_estime	gross annual household income
heures_travail_semaine	hours worked per week during studies
distance_domicile_campus_km	home to campus, km
premiere_generation_universitaire	1 if first in family to attend university

Column	Meaning
decision_octroi	target, 1 granted, 0 refused

Region values carry no accents and no spaces around the hyphen: **Montreal**, **Capitale-Nationale**, **Bas-Saint-Laurent**, **Cote-Nord**, **Gaspesie-Iles-de-la-Madeleine**.

Constraints

Fixed budget. Your grant rate on the 4,000 evaluation applicants must fall between 36% and 44%. Outside that range the technical section scores zero.

decision_octroi is not the target. Judges score against a reference standard built independently of the historical committee. You do not have it. The column records what the committee did, and the committee is under audit.

Deleting region_administrative does not work. Dropping it moves the parity gap from 0.188 to 0.181. Dropping the postal code as well moves it to 0.173. Distance, hours worked, household income and postal code all carry regional information. Section 5 of the notebook measures this.

Deliverables

A GitHub repository, public or shared with the judges, containing:

- **predictions.csv** at the root. Two columns, 4,000 rows plus a header, values 0 or 1. The last notebook cell writes a valid example.
- **audit_rapport.ipynb**. Measurement of the bias, your fairness metrics with justification, and the proxy variables you found.
- **model_corrige.py** or **.ipynb**. Your mitigation, with a Pareto front plot across several settings of the fairness constraint.
- **presentation.pdf**. Support for a five-minute pitch.

```
id_candidat,decision_octroi
C000 042,1
C000 117,0
```

Scoring

Section	Points	Judged by
Diagnostic rigour	25	jury
Technical solution	35	automated scorer
Governance and ethics	25	jury
Pitch and code quality	15	jury

The 35 automated points, both measured against the hidden reference standard:

- Equity, 20 points. Share of the baseline equal-opportunity gap closed. The baseline gap is 0.270.
- Utility, 15 points. Agreement with the reference standard, scaled between a random budget-respecting draw and a perfect allocation.

Both score zero if the budget constraint is broken.

Notes

fairlearn.postprocessing.ThresholdOptimizer adjusts decision thresholds after training and runs in seconds. **fairlearn.reductions.ExponentiatedGradient** retrains under a constraint and takes minutes.

Demographic parity and equal opportunity cannot both hold when the two groups have different profiles. Choose one and be ready to defend the choice.

A single model is not a Pareto front. Sweep the fairness constraint and plot the results.

Mentors are available throughout.